INCH-

MIL-G-24659A(SH)
AMENDMENT 1
17 April 1996

MILITARY SPECIFICATION

GAUGE, DIFFERENTIAL, DIAL INDICATING

This amendment forms a part of MIL-G-24659A(SH), dated 27 September 1993, and is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2: Above "Dial size", change "35" to "3" and add another designator at the right side of the page as follows:

" Y
Index
pointer
(see
1.2.11)"

1.2.1: Delete and substitute:

"1.2.1 Range. The differential pressure range shall be designated by two alpha-numeric characters. The first character shall represent the numerical base value of the range as specified in 3.3.2.2.3.1. The second character shall be the multiplier symbol and shall be limited to the following:"

PAGE 2

1.2.1: Add at the beginning of the list of multipliers: "R - multiply by 1"

1.2.4: Change the dial size designations "30, 35, 45, and 60" to "2, 3, 4, and 6" respectively.

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1.2.6: Delete the list of pressure ratings and designations and substitute:

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"1 - 500
2 - 1000
5 - 1500
6 - 2500
3 - 3000
4 - 6000"

PAGE 3

1.2.7: Add at the bottom of the list for material: "C - Beryllium copper (excluded for water application)"

1.2.8, Make the following changes to the fill fluid table:
Fill fluid class B: Delete "(68%)" and "(32%)." Change "1.08" to "1.02 - 1.08."
Fill fluid class I: Correct the spelling of "silicon" to "silicone." Change "#200-20" to "#200" and "0.95" to "0.87 - 1.02"
Fill fluid class S: Correct the spelling of "silicon" to "silicone." Add specific gravity "1.07."

PAGE 4

1.2.10: Add at the bottom of the list of material: "E - steel"

Add as new paragraph:

"1.2.11 Index pointer. The inclusion of an index pointer shall be designated by a single alpha character and shall be limited to the following (see 3.3.2.2.4):

Y - include an index pointer
N - do not include an index pointer"

2.1: Delete and substitute:

"2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or for examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed."

2.1.1: Under "**SPECIFICATIONS":**

SPECIFICATIONS

FEDERAL: Delete:
"QQ-A-591 - Aluminum Alloy Die Casting."

Add:
"TT-P-645 - Primer, Paint, Zinc-Molybdate, Alkyd Type."

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PAGE 5

MILITARY: Change "MILITARY" to "DEPARTMENT OF DEFENSE" and delete:

"MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-E-15090 - Enamel, Equipment, Light Gray (Formula No. 111).
MIL-P-83461 - Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C)."

Add:

"MIL-E-24635 - Enamel, Silicone Alkyd Copolymer (Metric).
MIL-C-24643/4 - Cord, Electrical, 600 Volts, Type LSMCOS."

Under "**STANDARDS**":

MILITARY: Change "MILITARY" to "DEPARTMENT OF DEFENSE" and add:

"MIL-STD-278 - Welding and Casting Standards."

PAGE 6

Add as new paragraph:

"2.1.2 Other Government documents, drawings and publications. The following other Government Documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 0900-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Application for copies should be addressed to the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)"

PAGES 6 AND 7

2.2 Under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)" delete:

"A 564 - Standard Specification for Hot-Rolled and Cold-Finished Age-Hardened Stainless and Heat-Resisting Steel Bars and Shapes.
A 581 - Standard Specification for Free-Machining Stainless and Heat-Resisting Steel Wire and Wire Rods. (Dod adopted) A 582 - Standard Specification for Free-Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished. (DoD adopted)"

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add:

"A 108 - Standard Specification for Steel Bars, Carbon Cold Finished, Standard Quality (AASHTO M169).
A 181/A 181M - Standard Specification for Forgings, Carbon Steel, for General Purpose Piping.
A 228/A 228M - Standard Specification for Steel Wire, Music Spring Quality.
A 313 - Standard Specification for Chromium-Nickel Stainless and Heat-Resisting Steel Spring Wire.
B 197 - Standard Specification for Copper- Beryllium Alloy Wire.
D 1974 - Standard Practice for Methods of Closing, Sealing, and Reinforcing Fiberboard Shipping Containers.
D 5118-D 5118M - Standard Practice for Fabrication of Fiberboard Shipping Boxes

PAGES 8 and 9

Table I: Delete the table and its notes and substitute:

"TABLE I. Indicating assembly materials.

Part	Material	Material specification	Remarks
Case	Aluminum 1/	ASTM B 26, ASTM B 85	2/
Window	Plastic	MIL-P-5425 ASTM D 3935	Lexan 300 series or equivalent
Movement	CRES	ASTM A 240, ASTM A 276	304L, 316L, 321, 347
Threaded fasteners	CRES Monel	ASTM A 276 QQ-N-286	304L, 316L, 321, 347 UNS N05500
Switches		MIL-S-8805	

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Electrical cable		MIL-C-24643/4	Low smoke
Electrical connector		MIL-C-5015	Both halves, see 3.3.2.2.2
Gaskets	Fluorocarbon rubber (250°F max) CRES Nickel Cellulose fiber	MIL-R-83248	Flat, type 2 Flat, model 199SS Flat, model 199SS Flat, model 199CS
O-rings	Fluorocarbon rubber Nitrile rubber (BUNA N) (180°F max)	MIL-R-83248 ASTM D 2000	

1/ The case shall be protected against corrosion by anodizing.
2/ The case finish shall include a primer in accordance with TT-P-645 and a grey enamel in accordance with MIL-E-24635."

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TABLE II: Delete and substitute:

"TABLE II. Sensing assembly materials.

Part	Material	Material specification	Remarks
Sensor and remote sensor	Inconel CRES Monel Beryllium copper	ASTM B 443 ASTM A 276 ASTM A 240 QQ-N-286 ASTM B 194	UNS N06625 304L, 316L, 321, 347 UNS S31603 UNS N05500
Range springs	Copper-nickel Elgiloy MP-35N Inconel CRES Beryllium copper Music wire	ASTM B 164 SAE AMS 5833 SAE AMS 5844 SAE AMS 5699 ASTM A 313 ASTM B 197 ASTM A 228	 UNS N07750 UNS S30200

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Gauge housing	Brass CRES Copper-nickel Monel Steel[1/]	ASTM B 21, ASTM B 124 ASTM A 276, ASTM A 479 MIL-C-15726 ASTM A 181/ASTM A 181M ASTM A 108	 C71500 (70-30) Grade 1018
Capillary tubing	Copper-nickel CRES	MIL-T-16420 ASTM A 213	715 (70-30) UNS S31600
Remote sensor housing	CRES Inconel Monel	ASTM A 276 ASTM A 240 QQ-N-281	304L, 316L, 318, 347 UNS S31600 Class A or B

1/ Steel housing shall be protected by a primer in accordance with TT-P-645 and a grey enamel in accordance with MIL-E-24635."

3.2.1: Delete and substitute:

"3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs."

PAGE 10

3.3.2.1.2: Third line: Change "figure 1" to "figures 1 and 2"

3.3.2.1.2.1: Delete the third sentence, "The tubing shall be....than 40 feet." Change the last sentence to read: "The tubing shall have a maximum of two union type joints, which shall be welded or brazed (see 3.3.5)."

3.3.2.2.1: First sentence: Change "an" to "a minimum" and "125" to "110."

PAGE 11

3.3.2.2.2: Third sentence: Change "two" to "one or two" and "MS3102E14S-6P" to "MS3102R14S-6P."

Seventh sentence: Change "MS3106E14S-6S" to "MS3106R14S-6S."
Eighth sentence: Change "MS3102E14S-12P" to "MS3102R24-19P."
Ninth sentence: Change "MS3106E14S-12S" to "MS3106R24-19S."

TABLE III: After "Range designation" add: "(see 1.2.1)" and before "Minor graduations" add "Minimum."

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3.3.2.2.4: Fourth sentence, delete and substitute: "An index pressure pointer shall be

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provided when specified (see 1.2.11 and 6.2)."

Add new paragraph:

"3.3.5 Welding and brazing. Remote sensing system tubing shall be joined by welding or brazing. Welding shall be in accordance with MIL-STD-278 and brazing shall be in accordance with NAVSHIPS 0900-001-7000."

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3.4.1.2 (a): Change "contacts" to "actuating mechanism."

3.4.1.2 (b): Change "contacts" to "actuating mechanism."

3.4.1.2 (c): Insert "contact" between "switch" and "actuation."

3.4.1.3.1: Change "1-1/2" to "1-3/4."

3.4.1.3.2 (a): Change "contacts" to "actuating mechanism" and "1-1/2" to "1-3/4."

3.4.1.3.2 (b): Change "contacts" to "actuating mechanism" and "2" to "2-1/4."

3.4.1.3.2 (c): Insert "contact" between "switch" and "actuation" and change "3-1/2" to "3-3/4."

3.4.1.3.2.1: Change "1-1/2" to "1-3/4."

PAGE 15

3.4.11: Add new sentence between second and third sentence:

"Switches are permitted a momentary change in state during any blow but must return to the original state after each momentary change in state."

PAGE 16

3.5: Delete in its entirety.

4.: Delete and substitute: "4. VERIFICATION."

4.1, 4.1.1, and 4.1.2: Delete in its entirety.

4.2 (b): Delete and substitute: "Conformance inspection (see 4.4)."

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4.4: Delete "Quality" in two places and capitalize "C" in "conformance" in title.

TABLE V: Delete "Quality" in two places and capitalize "C" in "conformance" in two places.

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PAGES 17 and 18

4.4.1: Delete "quality" in three places.

PAGE 22

4.7.12: Fifth sentence: Insert "a permanent" between "for" and "change."

4.8: Delete in its entirety.

5. through 5.1: Delete and substitute:

"5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity with the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

PAGE 23

6.2 (n): Delete.

PAGES 23 and 24

6.3: Delete in its entirety.

PAGES 24 and 25

6.7: Second sentence: Delete "military."

PAGE 25

Add as a new paragraph:

“6.10 Cross-reference. The classification (1.2) system has been changed in several areas. Cross-reference of the old and new classifications can easily be accomplished by correlating the following:

a. Dial size (1.2.4) -

Old	New	Dial size
30	2	3 inch
35	3	3-1/2 inch
45	4	4-1/2 inch

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60	6	6 inch

b. Pressure rating (1.2.6) -

Old	New	Pressure rating
1	1	500
2	2	1000
-	5	1500
-	6	2500
3	3	3000
4	4	6000

c. Sensor material (1.2.7) - added “C” for beryilluim copper.

d. Gauge housing (1.2.10) - added “E” for steel.

e. Index pointer (1.2.11) - this is a new requirement and appears as the last character in the classification.”

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FIGURE 1: Statement under title at top: Change "IS" in first line to "MAY BE." Change "1" inch dimension on top view to "1 1/8." Change "4" inch dimension and "3" inch dimension on the front view to "4 3/8" and "3 1/4" respectively. Change "2" inch dimension on the side view to "2 1/4."

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FIGURE 2: Statement under title at top: In the first line change "IS" to "MAY BE" and delete "3", 3 1/2",". Place an "*" before the "7 1/2, 7, and 7" inch dimensions for the front and side views.

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FIGURE 3: Delete the first line of the statement under the title at the top and replace with, "THE WIRING PORTS SHALL BE SEALED USING RECEPTACLE AND MATING PLUG IN ACCORDANCE WITH MIL-C-5015."
After "THE 3" INDICATOR" change "DOES" to "SHALL."

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FIGURE 4: For the "3" INDICATOR" change the "5 1/8" and "2 9/16" inch dimensions to "5 1/2" and "2 3/4" respectively. For the "3 1/2" INDICATOR" change the "5 1/8" and "2 9/16" inch dimensions to "5 1/2" and "2 3/4" respectively.

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FIGURE 5: Delete the title at the top of the page and substitute: "MOUNTING BRACKET DIMENSIONS." Delete the first line of the statement under the title at the top of the page and substitute: "THE MOUNTING BRACKET IS USED FOR 6" INDICATORS." Place an "*" before the "8" inch dimensions (three places). Add "SIDE VIEW" below the unlabeled view.

PAGES 30 THROUGH 40

APPENDIX A: Delete in its entirety.

Preparing activity:
Navy - SH
(Project 6685-N916)

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